

A literature-implied negative answer for unconditional frames in

$$C_r^*(F_2)$$

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2026-06-18

Status. This packet records a *literature-implied answer*, not a new proof packet. It gives a negative answer to a narrow question in Remark 2.2 of Rui Liu and Zhong-Jin Ruan, *Cb-frames for operator spaces*, arXiv:1601.06218: whether there is any unconditional frame for $C_r^*(F_2)$. It does not settle the broader cb-basis problem for separable C^* -algebras or weakly amenable reduced group C^* -algebras.

Source question. Liu–Ruan construct a cb-frame for $C_r^*(F_2)$ and observe in Remark 2.2 that their particular frame is not unconditional. The remark ends with the statement that, at that moment, it was not known whether there is any unconditional frame for $C_r^*(F_2)$.

Supporting theorem. Samir Kabbaj and Rafik Karkri, *Structure of Banach spaces and besselian Schauder frames*, arXiv:2411.00777, prove that an unconditional Schauder frame is Besselian and that a Banach space with a Besselian frame has Pełczyński’s property (u). They then state as a corollary that $C([0, 1])$ has no Besselian frame and consequently no unconditional frame.

Identification. Let a be a free generator of F_2 , and let $H = \langle a \rangle \cong \mathbb{Z}$. The canonical Fourier projection

$$E_H \left(\sum_{s \in F_2} c_s \lambda_s \right) = \sum_{s \in H} c_s \lambda_s$$

extends to the standard contractive conditional expectation from $C_r^*(F_2)$ onto $C_r^*(H)$. Hence $C_r^*(H)$ is a complemented Banach subspace of $C_r^*(F_2)$. Since $C_r^*(H) \cong C(\mathbb{T})$, and $C(\mathbb{T})$ is isomorphic to $C([0, 1])$ by the classical Milutin theorem, the supporting result rules out unconditional frames for $C_r^*(H)$.

Unconditional frames pass to complemented subspaces: applying a bounded projection to the frame vectors and restricting the coefficient functionals gives an unconditional frame for the complemented subspace. Therefore, if $C_r^*(F_2)$ had an unconditional frame, $C_r^*(H) \cong C(\mathbb{T})$ would also have one, contradicting the preceding paragraph.

Conclusion.

$C_r^*(F_2)$ has no unconditional frame.

Thus the unconditional-frame question in Liu–Ruan Remark 2.2 has a negative answer by later Banach-frame structure results.

Provenance and scope. The implication is agent-identified. The supporting paper does not appear to explicitly cite the Liu–Ruan question. The conclusion concerns ordinary Banach-space unconditional frames, matching the norm-unconditional convergence used in Liu–Ruan’s remark; it does not decide whether weakly amenable reduced group C^* -algebras have cb-bases.

References.

- R. Liu and Z.-J. Ruan, *Cb-frames for operator spaces*, arXiv:1601.06218.
- S. Kabbaj and R. Karkri, *Structure of Banach spaces and besselian Schauder frames*, arXiv:2411.00777.